CSE110: Programming Language I (Lab)
Fall 2025
Lab Exam - 03
Duration: 25 Minutes

A

Name:	ID:	Section:

Question 1 [CO2] [10 Points]

An ancient civilization called **Xylos** protected its interstellar systems using DNA-based locks. Each lock accepts a DNA string **S**, which contains only the characters **A, T, C, G,** and always has an even length between 6 and 50. To open the lock, the string must be decoded correctly.
The decoding is done **two characters (A block) at a time**, from left to right.

- **First character of each block**
  1. Normally, apply the complement rule: A ↔ T, C ↔ G
  2. If the previous decoded character was G, apply rotation instead: A→G, C→T, G→A, T→C (Trap rule)
- **Second character of each block**
  1. Normally, apply rotation: A→G, C→T, G→A, T→C
  2. If the first decoded character of the same block is A, apply the complement instead.
- Append the decoded characters in order to form the final string.

Print the final decoded DNA string.

Input	Output
ATCGAT	TCGATC
GATACA	CGCGTG

Test Case 2 Explanation:
Block 1 (G A):
First character G → Complement Rule → C (prev = none, normal complement)
Second character A → Rotation Rule → G
Result so far: CG

Block 2 (T A):
Previous decoded character = G → Trap Rule applies, first character uses Rotation instead of Complement T→ C
Second character A → Rotation Rule → G (first decoded ≠ A, normal rotation)
Result so far: CGCG

Block 3 (C A):
Previous decoded character = G → Trap Rule applies, first character uses Rotation
C → T
Second character A (first decoded = T → normal rotation) → G
Final decoded string: CGCGTG

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B

Name:	ID:	Section:

Question 1 [CO2] [10 Points]

A lost civilization called **Zyron** secured their space stations using DNA-based locks. Each lock takes a DNA string **S**, which contains only the letters **A, T, C, G** and always has an even length between 6 and 50. To unlock a station, the string must be decoded correctly.

The decoding is done **two characters at a time** from left to right.

- **First character of each block:**
  1. Normally, apply the complement rule: A ↔ T, C ↔ G
  2. If the previous decoded character was G, apply rotation instead: A→G, C→T, G→A, T→C (Trap Rule)
- **Second character of each block:**
  1. Normally, apply rotation: A→G, C→T, G→A, T→C
  2. If the first decoded character of the same block is A, apply complement instead (Mirror Rule)

Print the fully decoded DNA string

Input	Output
TACGGA	ATGACG

Explanation:

Block 1 (T A):
First character T → Complement Rule → A (prev = none, normal complement)
Second character A → Mirror Rule applies (first decoded = A) → T
Result so far: AT

Block 2 (C G):
Previous decoded character = T → Complement Rule → C→G
Second character G → Rotation Rule → A (first decoded ≠ A, normal rotation)
Result so far: ATGA

Block 3 (G A):
Previous decoded character = A → Complement Rule → G→C
Second character A → Rotation Rule → G (first decoded ≠ A, normal rotation)
Final decoded string: ATGACG

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In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A]

SOLUTION

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();

        String result = "";
        char prevDecoded = ' ';
        int i = 0;

        while (i < s.length()) {

            // First character
            char first = s.charAt(i);
            char decodedFirst;

            if (prevDecoded == 'G') { // rotation
                if (first == 'A') decodedFirst = 'G';
                else if (first == 'C') decodedFirst = 'T';
                else if (first == 'G') decodedFirst = 'A';
                else decodedFirst = 'C'; // T
            } else { // complement
                if (first == 'A') decodedFirst = 'T';
                else if (first == 'T') decodedFirst = 'A';
                else if (first == 'C') decodedFirst = 'G';
                else decodedFirst = 'C'; // G
            }

            result = result + decodedFirst;

            // Second character
            char second = s.charAt(i + 1);
            char decodedSecond;

            if (decodedFirst == 'A') { // complement
                if (second == 'A') decodedSecond = 'T';
                else if (second == 'T') decodedSecond = 'A';
                else if (second == 'C') decodedSecond = 'G';
                else decodedSecond = 'C'; // G
            } else { // rotation
                if (second == 'A') decodedSecond = 'G';
                else if (second == 'C') decodedSecond = 'T';
                else if (second == 'G') decodedSecond = 'A';
                else decodedSecond = 'C'; // T
            }

            result = result + decodedSecond;
            prevDecoded = decodedSecond;

            i = i + 2;
        }
        System.out.println(result);
    }}
```

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Rubric For Set A and Set B

SL	Points To Meet	Marks [10]
1	Correctly providing input and output and other things	2
2	Correctly implementing the complement rule for the first char	2
3	Correctly implementing the trap rule for the first char	2
4	Correctly implementing the rotation rule for the second char	2
6	Correctly implementing the complement rule for the second char	2

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[SET-B]

SOLUTION

```
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String s = sc.nextLine();
        String result = "";
        char prevDecoded = ' ';
        int i = 0;

        while (i < s.length()) {

            // First character
            char first = s.charAt(i);
            char decodedFirst;

            if (prevDecoded == 'G') { // Trap rule: rotation instead of complement
                if (first == 'A') decodedFirst = 'G';
                else if (first == 'C') decodedFirst = 'T';
                else if (first == 'G') decodedFirst = 'A';
                else decodedFirst = 'C'; // T
            } else { // Normal complement
                if (first == 'A') decodedFirst = 'T';
                else if (first == 'T') decodedFirst = 'A';
                else if (first == 'C') decodedFirst = 'G';
                else decodedFirst = 'C'; // G
            }

            result = result + decodedFirst;

            // Second character
            char second = s.charAt(i + 1);
            char decodedSecond;

            if (decodedFirst == 'A') { // Mirror rule: complement instead of rotation
                if (second == 'A') decodedSecond = 'T';
                else if (second == 'T') decodedSecond = 'A';
                else if (second == 'C') decodedSecond = 'G';
                else decodedSecond = 'C'; // G
            } else { // Normal rotation
                if (second == 'A') decodedSecond = 'G';
                else if (second == 'C') decodedSecond = 'T';
                else if (second == 'G') decodedSecond = 'A';
                else decodedSecond = 'C'; // T
            }

            result = result + decodedSecond;
            prevDecoded = decodedSecond;

            i = i + 2;
        }

        System.out.println(result);
    }
}
```

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